
Speaking Jewels into Spacetime: A Concept-and-Feasibility Study of Native Text-to-Video Gaussian Programs

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Abstract

1 Text-to-video systems usually generate a raster latent and decode it into frames. We
2 investigate a different target language: a video is cast directly as an irregular field of
3 persistent, anisotropic Gaussian Jewels in displayed spacetime (u, v, t) . Each Jewel
4 carries geometry, opacity, color, and a local color derivative; time-tilted support rep-
5 resents motion. The intended generator speaks a hierarchical executable program—
6 semantic scene and ownership tokens, trajectory phrases, physical Jewel tokens,
7 and continuous centroids—that an additive renderer turns into video. This paper
8 makes a bounded feasibility claim, not an open-vocabulary generation claim. In
9 controlled experiments, free spacetime tilt improves reconstruction in 9/9 matched
10 runs by 0.772 dB (95% CI [0.573, 0.971]); a support-correct amortized encoder
11 improves from 21.876 to 22.628 dB as training data grows from 12 to 120 videos;
12 and a prompt-and-seed-only exact compiler generates nine 72k-Jewel programs
13 with correct prompt similarity above shuffled and null controls in 9/9 cases and
14 OpenCLIP top-1 in 8/9. A 0.54M-parameter autoregressive speaker learns the
15 program syntax and recognizes all three held-out paraphrase classes, but fails the
16 stricter rendered retrieval gate. Current trajectory macros are source-backed. The
17 next experiment is to learn reusable, source-independent trajectory prototypes at
18 larger data and compute scale.

19 1 Introduction

20 A video can be treated as a stack of pixels, or as a continuous volume whose axes are horizontal
21 position, vertical position, and time. We ask whether a text-conditioned model can generate the
22 second object directly. Our primitive is a spacetime Gaussian Jewel: a colored, soft-support basis
23 function living in the displayed-video volume (u, v, t) . A slanted Jewel intersects consecutive time
24 slices at different image positions, so motion is part of its geometry rather than a correspondence
25 computed after frames are generated.

26 This target is neither a codec nor a claim of compactness. Video Gaussian representations have
27 already been studied for reconstruction, processing, and compression [16, 15, 18, 6, 14]. Our question
28 is different: can text select and compose persistent Gaussian operations as the *native output* of a
29 generative model? Such a model would speak into a continuous volume, keep explicit state across
30 temporal windows, and eventually permit local edits by modifying a bounded set of primitives.

31 The experiments separate representation capacity, semantic preservation, and native generation.
32 Reconstructing an existing video shows that the representation can hold video, not that language can
33 generate it. A prompt-conditioned scaffold followed by an encoder tests semantic preservation, not

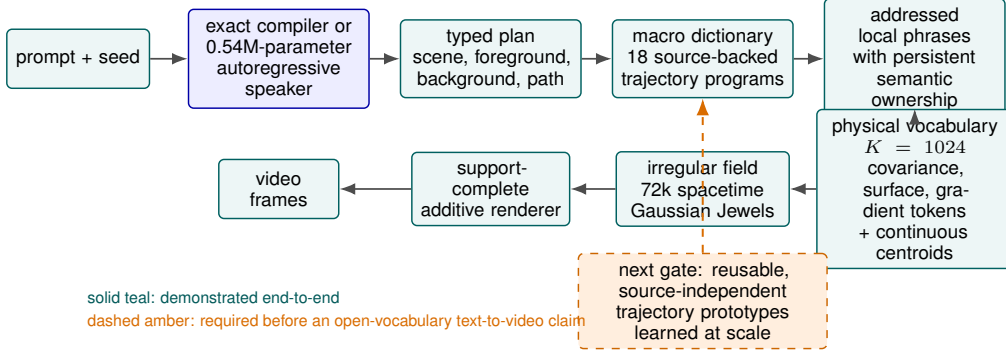


Figure 1: The proposed native text-to-video language. Solid boxes are demonstrated in this work; the dashed box is not. The exact compiler and learned speaker select a semantic plan, persistent foreground/background ownership, addressed phrases, and physical tokens. Current macro tokens are backed by 18 fitted sources.

native prompt-to-Jewel generation. Retrieval of a whole fitted source field is also not generation. We use input-ownership audits and shuffled/null controls to isolate the prompt-only claim.

Figure 1 shows both the demonstrated path and its missing component. At inference, the demonstrated prompt-only path receives exactly a prompt and integer seed, produces a typed plan, composes distinct foreground and background trajectory donors, expands local physical tokens at continuous centroids, and renders an irregular 72k-Jewel field. However, its macro dictionary is built from 18 fitted training fields. It does not retrieve a complete video, but it does reuse source-backed trajectory programs. The next gate replaces those macros with reusable, source-independent trajectory prototypes.

Our contributions are:

- a 22-parameter additive spacetime Gaussian primitive and a support-complete renderer that preserve continuous, irregular, time-tilted structure;
- causal and scaling evidence that temporal tilt improves reconstruction and that amortized video-to-Jewel prediction improves with data rather than collapsing to a coordinate grid;
- a hierarchical physical vocabulary with exact continuous centroids, three factorized $K = 1024$ token books, and measurable same-source canonicity;
- a bounded prompt-and-seed-only test of this pipeline, including exact and learned speakers, complete input audits, negative controls, and an explicit failure of the learned speaker’s strict rendered retrieval gate.

2 Related work

Dynamic and video Gaussians. 3D Gaussian Splatting represents radiance fields with explicit anisotropic primitives [5]. Dynamic extensions model time by deformation or native spacetime support [20, 7]. Video-oriented methods use Gaussians for processing, representation, or compression [16, 15, 18, 6, 14]. These works establish that Gaussians can represent changing visual signals. We focus on a displayed 2D video volume, additive rendering, and language-driven construction rather than novel-view synthesis or rate-distortion performance.

Generative structured media. Video diffusion generates dense latent tensors [3, 1]. Autoregressive multimodal models show that a language-modeling interface can emit visual tokens [2, 19], while Transfusion combines discrete language modeling with continuous visual generation [23]. Gaussian generation has been explored for 3D and dynamic scenes [21, 13, 8, 11]. Here, output tokens expand into persistent local operations in displayed spacetime. We test whether a hierarchical, executable Gaussian language has the required representational and semantic properties.

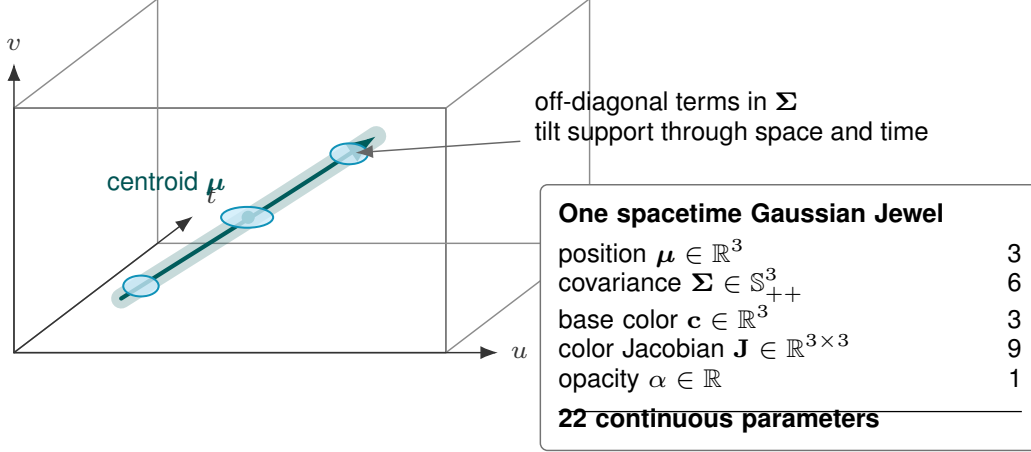


Figure 2: One spacetime Gaussian Jewel. A video frame is a fixed- t slice. The cross-section of tilted support changes position across slices, representing motion continuously. The color Jacobian permits local appearance change through space and time.

3 Method

3.1 A Jewel is a local function in displayed spacetime

Let the normalized video domain be $\Omega = [-1, 1]^3$, with query $\mathbf{x} = (u, v, t) \in \Omega$. Jewel i has center $\mu_i \in \mathbb{R}^3$, symmetric positive definite covariance $\Sigma_i \in \mathbb{R}^{3 \times 3}$, opacity logit ℓ_i , base RGB $\mathbf{c}_i \in \mathbb{R}^3$, and local color Jacobian $\mathbf{J}_i \in \mathbb{R}^{3 \times 3}$. We evaluate

$$q_i(\mathbf{x}) = (\mathbf{x} - \mu_i)^\top \Sigma_i^{-1} (\mathbf{x} - \mu_i), \quad (1)$$

$$a_i(\mathbf{x}) = \sigma(\ell_i) \exp[-q_i(\mathbf{x})/2], \quad (2)$$

$$\hat{\mathbf{I}}(\mathbf{x}) = \mathbf{b} + \sum_i a_i(\mathbf{x}) [\mathbf{c}_i + \mathbf{J}_i (\mathbf{x} - \mu_i)], \quad (3)$$

where $\mathbf{b}$ is a learned constant background. This is additive accumulation, not ordered alpha compositing. The primitive has 22 continuous parameters: 3 center, 6 covariance, 1 opacity, 3 base-color, and 9 first-order color parameters. Values outside the display range are clipped only for visualization; range violations are audited separately.

Writing $\Sigma_i = \mathbf{R}_i \text{diag}(\mathbf{s}_i^2) \mathbf{R}_i^\top$ makes motion geometric. Off-diagonal space-time covariance tilts support into t ; the intersection with successive fixed- t planes moves across (u, v) . Figure 2 illustrates this without the misleading picture of a flat frame translating laterally through image space.

3.2 Support-correct irregular rendering

The original fast renderer selected a fixed number of Euclidean nearest centers. That shortcut can silently omit wide or rotated Gaussians and confound a representation test with an approximation artifact. Our support-complete renderer tiles query voxels, finds every primitive whose exact Mahalanobis support lies within five standard deviations, and raises an error rather than dropping contributors when capacity is exceeded. At 10k, 45k, and 72k primitives it runs within roughly $2\times$ the k-nearest renderer; at 72k its measured peak allocation is 8.01 GB on an RTX 4090. Pixels and gradients agree with dense evaluation up to the declared support truncation.

Centroid placement is audited separately. Centroids are continuous and jittered; local phrases do not place one primitive on each fixed cell. We report the fraction of centers locked to the implied grid and require zero. This permits structured addressing without visible coordinate quantization.

3.3 A hierarchical executable language

A flat sequence of 72k independent 22-D records is too long and has no semantic ownership. We use four levels:

1. a *scene token* chooses the semantic family;
2. foreground and background *owner tokens* keep coherent donors along a full space–time trajectory instead of switching independently per cell;
3. addressed local phrases specify occupancy and which owner supplies each region; and
4. three physical token books independently quantize covariance, surface/base color, and color gradient, while centers remain continuous.

The factorization has 3×1024 active physical entries rather than a single combinatorial codebook. An eight-frame field requires 35,265 discrete decisions, below a preregistered 40k-token budget, and expands deterministically into an irregular field. The active representation reaches 22.866 dB token-only reconstruction PSNR, retains $1.042 \times$ the teacher’s median mixed space–time tilt, and has zero grid locking. Same-source physical token overlap is 0.452 versus 0.228 for different sources, a 0.224 canonicity margin.

In the tested prompt-only systems, trajectory-level ownership carried semantic structure. A global scene decoder and a purely addressed $16 \times 16 \times 8$ local phrase model each produced zero recognizable classes, although the local model improved token NLL from 6.123 to 5.168. Holding one coherent source through the volume recovered 3/3 classes. Assigning one foreground donor and a different background donor along a smoothed saliency trajectory preserved 3/3 and created a composition absent from any single source. Thus locality improves syntax, but trajectory-level ownership carries semantics.

3.4 Exact and learned speakers

The *exact compiler* maps three exact prompts and a seed to (scene, foreground owner, background owner, path). For each prompt, it chooses two distinct donors from six fitted fields in that class, with exactly 50/50 primitive ownership. The source choice is made once per temporal window, not independently per voxel. No target video, target field, block program, or held-out latent is provided at inference.

The *learned speaker* tests whether the plan has learnable syntax rather than only hand-written dispatch. A 541,223-parameter autoregressive network conditions on a 512-D text embedding and emits scene, foreground, then background tokens. It trains on 216 programs spanning each exact prompt and two paraphrases per class; an unseen fourth phrasing is reserved for evaluation. This is a small speaker over a closed three-scene vocabulary, not an LLM and not an open-vocabulary model.

4 Experimental design

We organize experiments as falsifiable gates rather than a cumulative demo. Every positive semantic claim includes a cyclic-shuffled prompt and a null-text control. Conditions share seeds and, where applicable, weights and targets. Prompt similarity is measured with OpenCLIP ViT-B/32 [12]; reconstruction uses PSNR and LPIPS [22]. CLIP is used as a screening metric, not as evidence of human-perceived video quality.

Causal representation test. We fit free full-covariance Jewels and a matched control whose space–time covariance is projected to zero after every update. Primitive count, parameter bytes, initialization family, and optimization budget are matched across three source videos and three seeds. We report paired differences and a two-sided Student- t 95% interval over the nine pairs.

Data scaling test. An amortized 5.26M-parameter encoder is trained on exact nested subsets of 12, 60, or 120 examples from five styles and 12 actions. Each budget has three independent seeds and is evaluated on the same frozen 60-video inventory. Models receive 120 matched corpus passes and a common low-rate continuation. Confidence intervals are t intervals over the three seed means; no arm is selected at a different exposure count. The fitted per-video teacher is an empirical ceiling, not a comparable generalizing model.

Prompt-only test. The exact compiler is evaluated on ballerina, dog, and welder prompts, three seeds each. Each output contains 72k Jewels and renders 49 frames at 144×216 . Success requires

Table 1: Evidence ledger. Pass means the declared bounded gate passed; it does not extend the scope stated in the final column. Exact per-run values are retained in machine-readable reports.

Question	Control	Result	Gate	What it establishes
Does time tilt help?	Full covariance vs. zeroed space-time terms; matched resources	Free wins 9/9; +0.772 dB, 95% CI [0.573, 0.971]	pass	Space-time covariance improves the tested reconstructions.
Does the encoder scale?	Nested 12/60/120 sets, 3 seeds, fixed 60-video validation	21.876 → 22.399 → 22.628 dB; LPIPS 0.473 → 0.465 → 0.392	pass	More data improves held-out prediction under matched protocol.
Is the physical language irregular and faithful?	Token-only decode; grid-lock and tilt audits	22.866 dB; zero grid lock; 1.042× tilt; 0.224 canonicity margin	pass	Factorized tokens preserve physical structure with exact centroids.
Can prompt + seed select video?	9 correct/shuffled/null programs; no target artifact at inference	Correct beats both controls 9/9; strict top-1 8/9; all class majorities	pass	A bounded source-backed native program is prompt-selective.
Can a small speaker generalize phrasing?	Unseen fourth paraphrase; frozen checkpoint; same render audit	Scene accuracy 3/3; NLL 2.201 vs. 4.387 shuffled; rendered top-1 4/9	partial	Program syntax is learnable; rendered open-ended generalization is not established.

that correct-prompt similarity exceed both shuffled and null controls for every program, with at least 6/9 strict OpenCLIP top-1 outcomes and majority success for all classes. The learned speaker uses the same render audit on unseen paraphrases; its weights are frozen at the best held-out correct-text NLL checkpoint before generating videos.

5 Results

5.1 Temporal tilt improves reconstruction

The free model beats the projected control in all nine paired fits. The mean paired PSNR gain is 0.77197 dB (sample standard deviation 0.25858 dB; 95% CI [0.57321, 0.97072]). Median mixed space-time tilt is 0.3459 in the free models and exactly zero by construction in controls. Gains are larger on higher-motion sources. Because primitive counts and serialized parameter bytes are matched, the result is evidence for the orientation of support, not extra capacity. It does not prove that every video benefits or that the current parameterization is optimal.

5.2 The amortized bottleneck improves with data

Figure 3 reports the predeclared scaling study. Mean held-out PSNR improves at both budget increases: 21.876 dB (95% CI [21.751, 22.000]), 22.399 dB ([22.344, 22.453]), and 22.628 dB ([22.459, 22.797]). Every 120-example seed beats every 60-example seed. From 60 to 120 examples, LPIPS falls 15.8%, layout PSNR rises from 30.726 to 31.533 dB, and median mixed tilt grows 2.58× from 0.0725 to 0.1875. The fitted teacher ceiling is 27.699 dB, LPIPS 0.1720, and median tilt 0.3966. Relative to the 120-example encoder, the measured gaps are 5.071 dB, 0.2196 LPIPS, and 0.2091 median tilt.

The same encoder preserves prompt information through a conservative text-to-pretrained-video-to-encoder route: correct prompt similarity is 0.3339 versus 0.1427 shuffled and 0.2096 null on 12 held-out actions. Correct beats shuffled in 12/12 cases and retains 91.4% of the source video’s prompt alignment. We exclude this result from the native-generation claim because a pretrained video model supplies the source video.

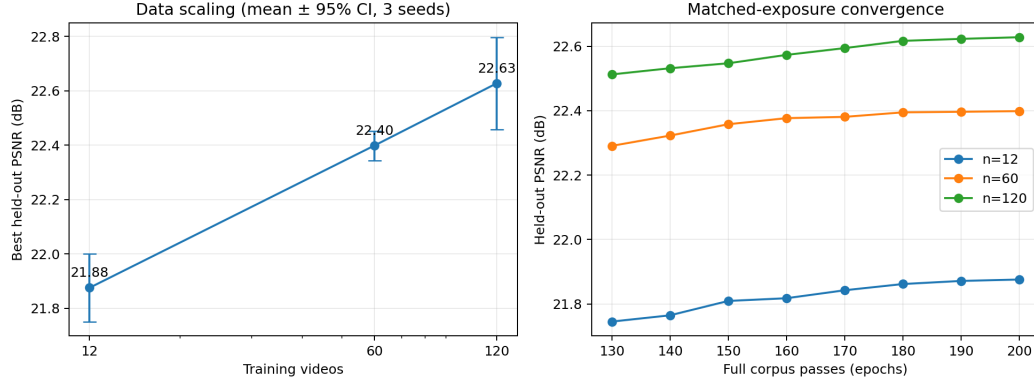


Figure 3: Support-correct encoder scaling. Left: held-out PSNR against training-set size, mean and 95% t interval over three seeds. Right: convergence against matched corpus exposure. The result is monotonic across these three small training sets.

5.3 The exact prompt-only compiler passes its bounded gate

Across nine programs, mean OpenCLIP similarity is 0.18173 for the correct prompt, 0.12525 for the cyclic-shuffled prompt, and 0.14825 for null text. The paired margins are $+0.05648$ and $+0.03348$; correct exceeds each control for all 9/9 programs. Strict top-1 is 8/9 and majority retrieval passes for all three classes. Foreground and background donor identifiers are distinct in every program, ownership is exactly balanced, and grid-lock fraction is zero. The input audit confirms that the only semantic inputs are exact prompt text and the declared integer seed.

Figure 4 shows all conditions rather than selected successes. The exact compiler is visibly class-selective and composes donor regions, but quality is far below a modern dense video generator. Because the macro vocabulary is source-backed, this result establishes execution and semantic selection, not novel open-vocabulary motion.

5.4 Learned speaker results

The learned speaker reaches its best held-out correct-text NLL at step 100 and continues to step 1100 before the declared plateau rule stops training. On unseen paraphrases, macro token NLL is 2.2005 for correct text, 4.3870 for shuffled text, and 2.5391 for null text. Scene accuracy is 100% for correct text, 0% shuffled, and 33% null; all nine correct-text samples are scene-consistent. In rendered space, correct similarity exceeds shuffled by 0.04782 on average and null by 0.04117, with 7/9 correct-over-shuffled pairs.

The strict rendered top-1 test passes only 4/9, below the declared 6/9 threshold, and only one of three class majorities passes. The network learned the coarse scene syntax under unseen wording but did not reliably choose donor combinations whose low-fidelity renders win a frozen semantic retrieval test. Held-out correct NLL plateaued early, so the next experiment changes the vocabulary and supervision rather than extending the same training run: learn reusable trajectory prototypes from many sources, train the speaker against rendered semantic and temporal objectives, and compare against a parameter-matched direct continuous head.

6 Scaling priorities

The experiments isolate three different scaling questions.

Representation compute. Longer local appearance optimization improves the frozen-geometry derivative head. For one seed, LPIPS falls from 0.717 at 400 updates to 0.670 at 4k and 0.659 at 12k; a second seed reaches 0.660 at 12k. The final seeds differ by only 0.00058 LPIPS and 0.00041 dB PSNR, and geometry remains bit-identical. The plateau after roughly 8–12k updates limits the expected gain from extending the same optimization run.

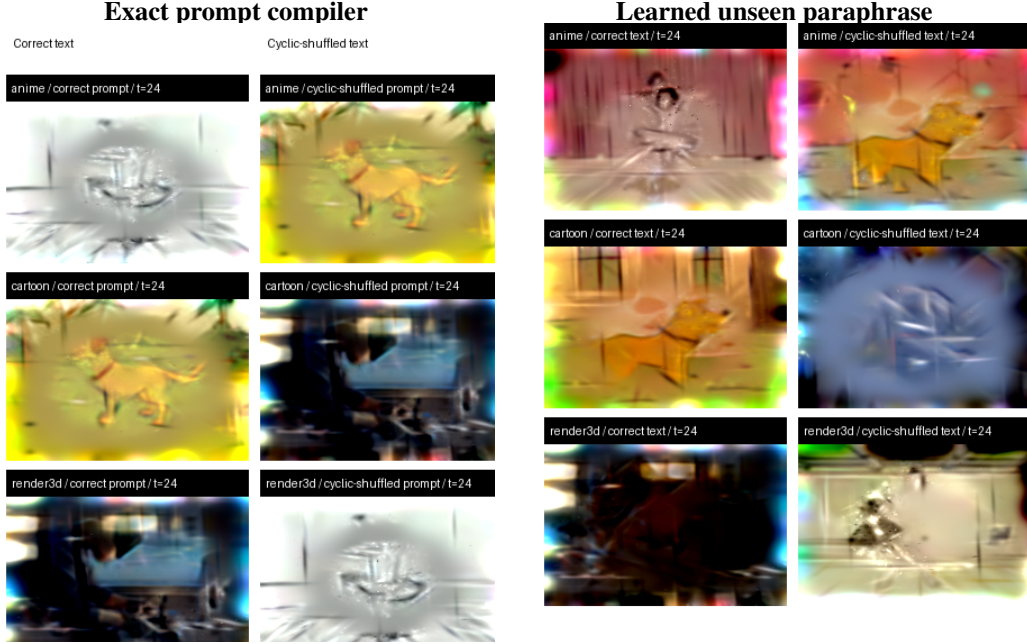


Figure 4: Prompt-only evidence sheet reflowed from one audited image. Each half shows correct-prompt and cyclic-shuffled generation for ballerina, dog, and welder classes. All evaluated samples are shown. These are 72k-Jewel renders, not source frames; their trajectory macro tokens are nevertheless backed by fitted training fields.

196 **Encoder data.** Held-out PSNR rises at each data scale and remains below the fitted teacher. The
 197 current 120-example maximum is far too small for a foundation model claim. A follow-up study
 198 should fix architecture and corpus passes, increase data by at least another order of magnitude, and
 199 measure perceptual fidelity, layout, mixed tilt, and prompt retention jointly.

200 **Generative vocabulary.** We propose segmenting many fitted fields into persistent trajectory tubes
 201 using motion and semantic features, learning a dictionary whose entries aggregate across source
 202 videos, and holding out entire motions, appearances, and compositions. A speaker should emit
 203 prototype identities plus continuous transforms and residuals; a local flow or diffusion head can
 204 model residual 22-D marks [9, 10, 4]. The proposed gate requires prompt-only renders whose macro
 205 tokens are source-independent, whose prompts and motions are held out compositionally, and whose
 206 correct/shuffled/null margins persist over many classes and seeds.

207 7 Limitations

208 **Closed semantic domain.** The prompt test contains three classes and 18 source fields. It does not
 209 measure open vocabulary, long-form prompting, camera control, or physical interaction.

210 **Source-backed macro vocabulary.** Foreground and background trajectories reuse programs derived
 211 from fitted sources. Although no whole video is retrieved and two donors are composed, the system
 212 has not learned source-independent motion concepts. This prevents an open-vocabulary claim.

213 **Small, synthetic evaluation.** OpenCLIP retrieval is noisy at the current visual quality and does not
 214 replace human judgment, FVD [17], or a broad action benchmark. The reported confidence intervals
 215 cover seeds within fixed small datasets, not dataset uncertainty.

216 **Rendering and color.** Additive overlap can wash out or overshoot RGB range, and the renderer
 217 does not model occlusion through ordered alpha compositing. Support truncation is exact only with
 218 respect to the declared five-sigma approximation.

219 **No demonstrated efficiency advantage.** We do not claim that a Jewel video is smaller, cheaper, or
 220 higher quality than a latent video. A 72k-by-22 continuous field is not intrinsically a compression

format. The motivation is persistent executable structure; editing and scaling advantages remain hypotheses requiring direct baselines.

Negative experiments are local evidence. Failure of a particular global decoder, local phrase model, or learned retrieval gate identifies a missing dependency under its tested data and optimizer. A single failed run is not treated as a law about Gaussian languages.

8 Broader impact

An explicit spacetime program could enable inspectable animation tools, localized corrections, and reuse of persistent state. If future models make local edits without regenerating an entire video, they may reduce iteration cost. The same capability can enable deceptive synthetic media, impersonation, or biased visual depictions; explicit primitives do not remove those harms. Before releasing a capable open-ended generator, appropriate measures include provenance metadata, watermarking where robust, prompt and output abuse screening, consent-aware data curation, bias and identity audits, and staged access. Training and rendering also consume energy. Efficiency claims should be based on measured end-to-end quality-matched compute, not primitive counts alone.

9 Conclusion

The measurements support a specific proposition: persistent time-tilted Gaussian Jewels improve reconstruction under matched resources, their amortized prediction improves with data, their physical parameters admit an irregular hierarchical tokenization, and prompt plus seed can drive a bounded native program with controlled semantic selectivity. They do not support the claim that we have a general text-to-video model. The next step is to replace source-backed trajectory macros with reusable learned prototypes and test compositional prompt-only generation on held-out motions and appearances. That experiment tests whether the current result scales with additional data and compute.

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A Protocols and preregistered gates

Temporal-tilt gate. The inventory contains three source videos and seeds 0, 1, and 2. Free and projected models share primitive and serialized parameter counts. The projection zeroes covariance entries (u, t) and (v, t) after each optimizer step. The gate requires a positive mean paired PSNR difference, a 95% interval excluding zero, and removal of measured mixed tilt in the control.

Encoder-scaling gate. Nested train sets contain 12, 60, and 120 examples. The common validation set contains 60 examples and has SHA-256 digest 3e034a1263e21b566c79b63a8257e9794c4c66316e3a82d6fe65d418d1699b25. The encoder uses 4,096 sampled query points per step, AdamW, an initial matched 120-corpus-pass phase, and a continuation at learning rate 3×10^{-5} for at least 40 and at most 80 additional passes. Evaluation occurs every 10 passes, with patience 3 and minimum improvement 0.03 dB. The reported endpoint is the common continuation protocol, not per-arm cherry-picked training length.

Physical-language gate. The active books contain 1024 entries for covariance, 1024 for base/surface color, and 1024 for the color Jacobian. Centers are not quantized. Required checks include finite metrics, token-only PSNR above 20 dB, mixed-tilt retention above 0.8, exactly zero grid locking, fewer than 40k decisions for eight frames, and same-source canonicity margin above 0.05.

Prompt-only gate. The exact prompts are a ballerina spinning a pirouette in a studio, a golden retriever catching a ball on grass, and a welder joining steel with bright sparks. Seeds are 20260914, 20260915, and 20260916. Controls cyclically rotate prompts or remove text while holding the declared seed. The learned speaker trains on each exact prompt and two paraphrases; a fourth paraphrase per class is held out. Its AdamW learning rate is 0.002, batch size 64, text-dropout probability 0.1, minimum 1000 steps, maximum 5000 steps, evaluation interval 100, and plateau patience 10 evaluations with minimum NLL improvement 10^{-4} .

B Additional quantitative detail

Table 2: Encoder scaling audit. Intervals are 95% t intervals over three seed means.

Train videos	PSNR (dB)	95% CI	LPIPS	Layout PSNR
12	21.876	[21.751, 22.000]	0.4728	29.812
60	22.399	[22.344, 22.453]	0.4651	30.726
120	22.628	[22.459, 22.797]	0.3916	31.533
Teacher	27.699	—	0.1720	—

The learned speaker’s higher absolute similarity is not evidence that it passes the stricter gate: the relevant outcome is relative retrieval among candidate prompts for each generated sample. It passes correct-over-shuffled in 7/9 cases but strict top-1 in only 4/9.

Table 3: Prompt-only semantic audit. Similarities are OpenCLIP cosine values.

Speaker	Correct	Shuffled	Null
Exact compiler	0.18173	0.12525	0.14825
Learned, unseen paraphrases	0.25056	0.20274	0.20939

C Compute and implementation

Experiments ran on local NVIDIA RTX 4090 (24 GB) and RTX 2070 Super (8 GB) GPUs. Support-complete 72k-field rendering measured 8.01 GB peak allocation on the RTX 4090; chunked paths permit 45k–120k field fitting on the 2070 Super. The encoder has 5.26M trainable parameters. Representative 12-example encoder continuations take approximately 129 seconds for 2,400 updates on the recorded machine; larger arms scale with corpus size and evaluation cost. The learned speaker has 541,223 parameters and trains for 1,100 updates in the selected run. Derivative appearance studies run to 12k–16k updates. Total exploratory compute exceeds these selected runs because architectural screening, renderer validation, and failed controls are retained in the repository rather than omitted. A submission artifact should record exact GPU model, driver, package lock, wall time, and energy estimate for every reproduced run.

D Reproducibility map

The following repository-relative machine-readable reports support the manuscript:

- `sol/results/temporal_tilt_replication_v1/report.json`;
- `sol/results/encoder_convergence_v2_continued/aggregate/report.json` and `audit/audit.json`;
- `sol/results/jewel_casting_language_v0/hierarchical_v1/gate0f_individual/report.json`;
- `sol/results/jewel_casting_language_v0/prompt_trajectory_speaker_v1/report.json`;
- `sol/results/jewel_casting_language_v0/learned_trajectory_speaker_v1/report.json` and `audit/report.json`; and
- `sol/results/local_adapter_convergence_v1/README.md`.

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